Air Power:

Ground Unit and Ground Target Data

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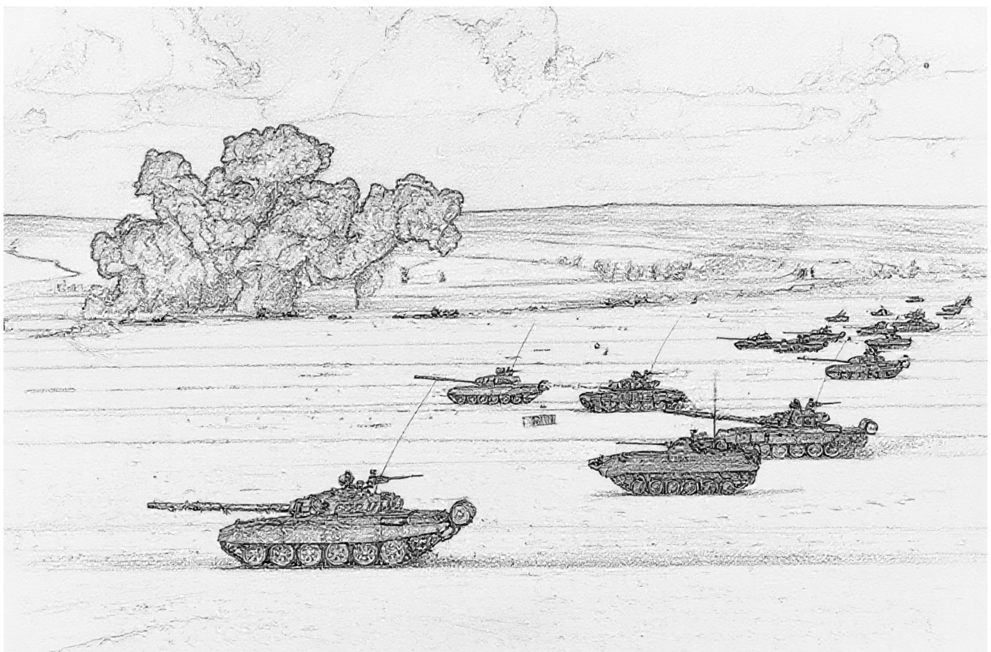


INTRODUCTION

For many, the principal focus of the *Air Power* game is air-to-air combat. That said, no less an authority on air-to-air combat than Robin Olds stated:[1]

> "You're not going to win a war by shooting down a damn MiG, you're going to win a war by bombing the bejesus out of whatever you were sent to bomb."

Here we deal with those "whatevers," the ground units and ground targets that are the targets of air-to-ground attacks, and some of which shoot back.

This document first presents ground-combat and transport units, then air-defense units (AAA, FCR, SAM, EWR, and CCU units), and finally ground targets. An appendix gives some examples of battalion-sized units.

Much of the data here is adapted from *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*, from *Air Power Journal*, and from Malcolm Pipe's extensive compendium of AAA and SAM units. However, the data have been carefully revised to improve their accuracy and internal consistency, a wider variety of ground-combat units are covered to give a richer experience, and the descriptions of the units and their weapon systems have been expanded.

Extensive notes accompany the main text, providing provenance of data, explanations of changes, sources of information, and justifications for decisions.

CONTENTS

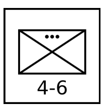
GROUND-COMBAT AND TRANSPORT UNITS

Ground-combat units are formations of infantry, tanks, IFVs, APCs, other AFVs, or artillery whose primary role is combat with other ground units. Transport units provide strategic mobility to infantry and towed artillery and are the life-blood of logistic networks. These units are presented below with their counters. The Ground-Combat and Transport Unit Table summarizes their properties. Some representative examples of vehicles are given in the descriptions, and Vehicle Types Table gives a more extensive list. All ground-combat and transport platoons and batteries are also available as sections and squads.[2]

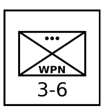
INFANTRY UNITS

Infantry are soldiers trained to fight on foot and, in some cases, from IFVs. Their primary roles in modern warfare are seizing, clearing, and holding ground, often as part of a combined-arms formation with tanks and artillery, highly mobile offensive and defensive operations in combination with helicopters, and combat in difficult terrain, including urban areas, mountainous zones, and dense jungle or forest.

An important secondary characteristic of infantry is any associated transport. Motorized infantry units are transported by trucks, giving them operational mobility. Mechanized infantry are transported by APCs or IFVs, giving them both tactical and operational mobility. Light infantry units have no or fewer supporting vehicles, although modern light infantry is often airborne.

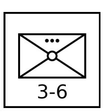


Infantry Platoon: An infantry platoon is armed principally with small arms, often supplemented with light machine guns and sometimes with light grenade launchers and portable anti-tank weapons such as bazookas, LAWs, MAWs, and portable ATGMs. It is the primary combat unit in larger infantry formations.



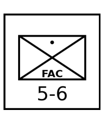
Infantry Weapons Platoon: An infantry weapons platoon is equipped with heavier weapons, such as medium or heavy machine guns, recoilless rifles, portable ATGMs, and automatic grenade launchers.

It is often a company- or battalion-level asset, and its role is to provide direct-fire support to the infantry platoons in its formation. [3]



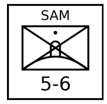
Infantry Mortar Platoon: An infantry mortar platoon is equipped with 60-mm, 81-mm, 82-mm, or similar mortars. It is are often battalion-level assets, and its role is to provide indirect fire support and smokescreens to the infantry platoons in its formation.

Due to the weight of ammunition required for sustained fire, it will often be transported by a light truck platoon even if the other infantry units in its formation do not have transport.



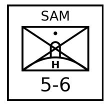
Infantry FAQ Squad: An infantry FAC squad has a forward air controller with other soldiers providing communication and protection.

An infantry FAC squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



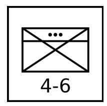
Infantry SAM Squad: Infantry SAM squads are equipped with shoulder-launched SAMs, with one launcher per squad. Their SAM capabilities and VP values are given elsewhere.

An infantry SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.



Infantry Heavy SAM Squad: Infantry heavy SAM squads are equipped with pedestal-mounted SAMs, with one launcher per squad. Their SAM capabilities and VP values are given in a subsequent section.

An infantry heavy SAM squad will often form a composite platoon with an infantry or infantry weapons platoon. If operating independently, it will often be transported by a light truck squad.

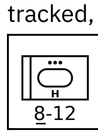


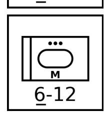
Infantry HQ Platoon: An infantry HQ platoon is the HQ unit of an infantry battalion or regiment. It consists of the formation commander, their staff, and their protection detail.[4]

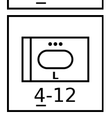
ARMORED VEHICLE UNITS

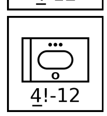
Armored ground-combat vehicle units combine armored protection for their crew and weapons, mobility, and either a firepower or transport capability.

Armored cars and armored half-tracks are considered to be light tanks, light anti-tank missile vehicles, light reconnaissance vehicles, light IFVs, or light or open APCs according to their role and armament. That is, the means of traction — tracked, half-tracked, or wheeled — is a secondary consideration.









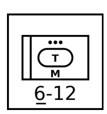
Tank Platoon: A tank platoon has armored vehicles principally armed with a direct-fire gun. The gun is most commonly mounted in a turret, but AFVs with casemate guns are also considered to be tanks. Its role is the support of infantry and the destruction of other tanks and armored vehicles. A tank platoon may have heavy, medium, light, or open armor. [5]

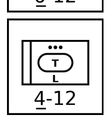
Heavy examples: M1 Abrams, Chieftain, Challenger, Leopard 2, and T-64/72/80.

Medium examples: M60, Centurion, Leopard 1, and T-54/55/62.

Light examples: M114A2, M551 Sheridan, Fox, Scimitar, Scorpion, and PT-76.

Open example: ASU-57.

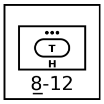


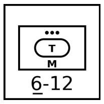


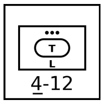
IFV Platoon: An infantry fighting vehicle (IFV) platoon is equipped with armored vehicles that transport infantry and provide them with fire support, being armed with a 20 mm or larger cannon and sometimes an anti-tank missile. Some IFVs are also designed to allow their infantry to fight while mounted. An IFV platoon may have medium or light armor.[6]

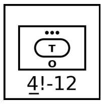
Medium example: M2A2 Bradley and Marder.

Light examples: M2/M2A1 Bradley, FV432 RARDEN, Warrior, and BMP-1/2.

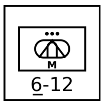


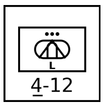




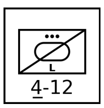


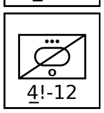
APC Platoon: An APC platoon is equipped with armored vehicles that transport infantry, but unlike IFVs are not equipped to provide fire support and at most are armed with a heavy machine gun. An APC platoon may have heavy, medium, light, or open armor. Heavy and medium APCs are typically converted tanks.[7]
Heavy examples: Nakpadon.
Medium examples: Nagmashot and Nagmachon.
Light examples: M113, FV432, Spartan, and BTR-50PK/60PB/70.
Open examples: M3 half-track and BTR-50P/60P.



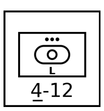


Armored Anti-Tank Missile Platoon: An armored anti-tank missile platoon has armored vehicles equipped with anti-tank guided missiles. Its mission is the destruction of other armored vehicles. Conventional gun-armed tank destroyers such as the Kanonen Jpz are considered to be tanks. It is often a battalion- or regiment-level asset. An anti-tank platoon may have medium or light armor.[8]
Medium examples: Rakaten Jpz and Jaguar.
Light examples: M150, M901, FV438, Striker, and BRDM.

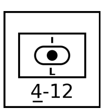


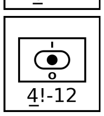


Armored Reconnaissance Platoon: An armored reconnaissance platoon has armored vehicles specialized in scouting and observation and with no offensive weapons more powerful than machine guns. (Vehicles used for reconnaissance that do have significant offensive weapons are typically classified as light tanks or IFVs.) A reconnaissance platoon may have light or open armor.[9]
Light examples: M114, Ferret, and BRDM-2.
Open example: M20.

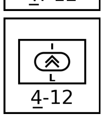


Armored Mortar Platoon: An armored mortar platoon has mortars mounted in armored vehicles (most commonly converted APCs or IFVs). It is typically a battalion-level asset in mechanized-infantry or tank battalions, and its role is indirect fire support. An armored mortar platoon has light armor.[10]
Examples: M106 and M125.

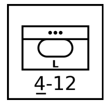




Armored Artillery Battery: An armored artillery battery has guns or howitzers in armored vehicles. It is typically a regiment-, brigade-, or division-level asset, and its role is indirect fire support. An armored artillery battery may have light or open armor.[11]
Examples: M109, 2S3, and 2S1.



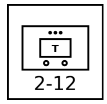
Armored Rocket Artillery Battery: An armored rocket artillery battery is the rocket-armed equivalent of an armored artillery battery. An armored rocket artillery battery has light armor.[12]
Example: M270.



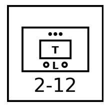
Armored HQ Platoon: An armored HQ platoon is the HQ unit of a tank, mechanized-infantry, or armored-artillery battalion, regiment, or brigade. It consists of the formation commander, their staff, and their protection detail. It is typically equipped with modified APCs or IFVs. An armored HQ platoon has light armor.

UNARMORED VEHICLE UNITS

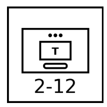
Unarmored or soft vehicles provide mobility, but do not provide (or provide only incomplete) armor protection for their crew and weapons. They include armored vehicles whose crew or weapons are exposed. They may be wheeled or tracked.



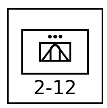
Truck Platoon: A truck platoon has unarmored wheeled vehicles that can transport medium or heavy loads of infantry, towed weapons, and supplies, including fuel and ammunition.



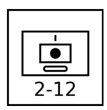
Light Truck Platoon: A light truck platoon has unarmored wheeled vehicles that can transport light loads of infantry, towed weapons, and supplies, including fuel and ammunition.
Examples: M151 Jeep, HMMWV, Landrover, Ilitis, and GAZ-69.



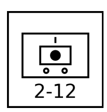
Tracked Carrier Platoon: A tracked carrier platoon is equipped with unarmored tracked vehicles that can transport medium loads of infantry, towed weapons, and supplies, including fuel and ammunition.[13]
Example: Bv 202.



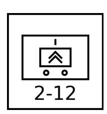
Mobile Anti-Tank Missile Platoon: A mobile anti-tank missile platoon is equipped with anti-tank missiles mounted on unarmored light trucks.
Examples: M151 TOW and M966 HMMWV TOW.



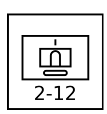
Tracked Artillery Battery: A tracked artillery battery has guns or howitzers mounted on tracked vehicles that provide mobility, but give only limited protection from small arms or artillery shrapnel. It is typically a regiment-, brigade-, or division-level asset, and its role is indirect fire support.[14]
Examples: M107, M110, 2S4, and 2S7.



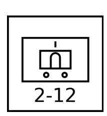
Mobile Artillery Battery: A mobile artillery battery has howitzers mounted on wheeled vehicles that provide mobility, but give only limited protection from small arms or artillery shrapnel. It is typically a regiment-, brigade-, or division-level asset, and its role is indirect fire support.[15]
Examples: CAESAR and Archer.



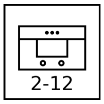
Mobile Rocket Artillery Battery: A mobile rocket artillery battery is the rocket-armed equivalent of a mobile artillery battery.[16]
Examples: M142 HIMARS and BM-21 Grad.



Tracked Missile Artillery Battery: A tracked rocket artillery battery is the rocket-armed equivalent of a mobile tracked battery.[17]
Example: M752 Lance.



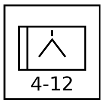
Mobile Missile Artillery Battery: A mobile missile artillery battery is the missile-armed equivalent of a mobile artillery battery.[18]
Examples: M1003 Pershing II and OTR-21 Tocha (SS-21).



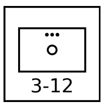
Mobile HQ Platoon: A mobile HQ platoon is typically the HQ unit of a battalion, regiment, or brigade other than those of infantry, tank, mechanized infantry, or armored artillery. It is equipped with trucks, and its personnel are not specialized in fighting as infantry.

TOWED UNITS

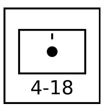
Towed weapons provides neither inherent mobility nor protection. However, they are light and cheap compared to similar vehicle-mounted weapons. Towed weapons is typically used away from direct combat and provided with appropriate ground or helicopter transport.



Towed Anti-Tank Gun Battery: A towed anti-tank gun battery has anti-tank guns on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset.
Example: 2A18 (T-12) and 2A45 (Sprut).



Towed Mortar Platoon: A towed mortar platoon has 120 to 160 mm mortars on unprotected, towed mounts. It is often a regimental- or brigade- asset. Smaller infantry mortars are found in infantry mortar platoons.
Example: M1943.



Towed Artillery Battery: A towed artillery battery has guns or howitzers on towed mounts with at most only very limited protection. It is often a regimental-, brigade-, or divisional-level asset.
Examples: M119, M198, and 2A18 (D-30).

Ground-Combat and Transport Unit Table

Type	Size[a]	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	AAA Class[b]
Infantry	Platoon	4	6	U	5/3/2	B2
Infantry Weapons	Platoon	3	6	U	6/4/2	B2
Infantry Mortar	Platoon	3	6	U	6/4/2	B2
Infantry FAC	Squad	5	6	U	–/–/8	—
Infantry SAM	Squad	5	6	U		—
Infantry Heavy SAM	Squad	5	6	U		—
Infantry HQ	Platoon	4	6	U	10/7/3	B2
Heavy Tank	Platoon	8	12	U	12/8/4	B3
Medium Tank	Platoon	6	12	U	10/7/3	B3
Light Tank	Platoon	4	12	U	6/4/2	B3
Open Tank	Platoon	4!	12	U	5/3/2	B3
Medium IFV	Platoon	6	12	U	10/7/3	B3
Light IFV	Platoon	4	12	U	6/4/2	B3
Heavy APC	Platoon	8	12	U	10/7/3	B3
Medium APC	Platoon	6	12	U	8/5/3	B3
Light APC	Platoon	4	12	U	6/4/2	B3
Open APC	Platoon	4!	12	U	5/3/2	B3
Medium Armored Anti-Tank Missile	Platoon	6	12	U	12/8/4	B3
Light Armored Anti-Tank Missile	Platoon	4	12	U	10/7/3	B3
Light Armored Reconnaissance	Platoon	4	12	U	6/4/2	B3
Open Armored Reconnaissance	Platoon	4!	12	U	4/3/1	B3
Light Armored Mortar	Platoon	4	12	U	10/7/3	B3
Light Armored Artillery	Battery	4	12	U	12/8/4	B3
Open Armored Artillery	Battery	4!	12	U	12/8/4	B3
Light Armored Rocket Artillery	Battery	4	12	U	12/8/4	B3
Light Armored HQ	Platoon	4	12	U	12/8/4	B3
Truck	Platoon	2	12	G/P	4/3/1	—
Light Truck	Platoon	2	12	G/P	4/3/1	—
Tracked Carrier	Platoon	2	12	U	4/3/1	—
Mobile Anti-Tank Missile	Platoon	2	12	G/P	8/5/3	—
Mobile Artillery	Battery	2	12	G/P	9/6/3	—
Tracked Artillery	Battery	2	12	U	10/7/3	—
Mobile Rocket Artillery	Battery	2	12	G/P	9/6/3	—
Tracked Missile Artillery	Battery	2	12	U	10/7/3	—
Mobile Missile Artillery	Battery	2	12	G/P	9/6/3	—
Mobile HQ	Platoon	2	12	G/P	10/7/3	—
Towed Mortar	Platoon	3	12	T	7/5/2	—
Towed Anti-Tank Gun	Battery	4	12	T	8/5/3	—
Towed Artillery	Battery	4	18	T	8/5/3	—

[a] Platoons and batteries are available as sections and squads with corresponding modifications to their damage resiliences and VPs. [b] Platoons with AAA class of B2 and B3 can employ barrage fire to altitudes of 2 and 3 levels, respectively.

Vehicle Types Table

Heavy Tanks		
Al-Khalid/VT-1A	PK/CN	2001
Ariete	IT	1995
Arjun	IN	2004
Challenger 1	GB	1983
Challenger 2	GB	1998
Chieftain	GB	1967
Conqueror	GB	1955
IS-2	SU	1944
IS-3	SU	1945
K1	KR	1987
K2 Black Panther	KR	2014
Leclerc	FR	1992
Leopard 2	DE	1979
M-84	YU	1985
M1 Abrams	US	1980
Merkava 1	IL	1979
Merkava 2	IL	1983
Merkava 3	IL	1989
Merkava 4	IL	2004
PT-91	PL	1995
Stridsvagn 103	SE	1967
T-10	SU	1954
T-14 Armata	RU	2024
T-64	SU	1966
T-72	SU	1973
T-80	SU	1976
T-84	UA	1999
T-90	RU	1992
Type 10	JP	2012
Type 90	JP	1990
Type 99	CN	2001

Medium Tanks		
AMX-30	FR	1966
ASU-85	SU	1959
Centurion	GB	1945
Comet	GB	1944
Cromwell	GB	1944
ISU-122	SU	1944
ISU-152	SU	1943
Kanonen JPz	DE	1965
Leopard 1	DE	1965
M-50 Sherman	IL/FR	1956
M-51 Sherman	IL/FR	1961
M103	US	1957
M26 Pershing	US	1944
M4 Sherman	US	1942
M46 Patton	US	1949
M47 Patton	US	1951
M48 Patton	US	1952
M60	US	1959
Panzer 61	CH	1965
Panzer 68	CH	1971
SU-100	SU	1944
T-34	SU	1940
T-54	SU	1955
T-55	SU	1958
T-62	SU	1961
TAM	AR	1983
Type 59	CN	1959
Type 61	JP	1961
Type 69	CN	1972
Type 74	JP	1975
Type 79	CN	1984
Type 85	CN	1988
Type 88	CN	1988
Type 96	CN	1997
Type 15	CN	2018
Vickers MBT	GB	1965
Vijayanta	IN/GB	1965

Light Tanks		
AMX-10 RC	FR	1981
AMX-13	FR	1953
Fox	GB	1975
Commando (90 mm)	US	1964
LAV I Cougar	CA	1976
LAV II Coyote	CA	1990
M8/Greyhound	US	1943
M24 Chaffee	US	1944
M41 Walker Bulldog	US	1953
M113 C&V with 25 mm	US	1974
M114A2	US	1969
M551 Sheridan	US	1969
Panhard AML	FR	1961
Panhard EBR	FR	1951
PT-76	SU	1951
Saladin	GB	1958
Scimitar	GB	1974
Scorpion	GB	1973
SpPz Luchs	DE	1975
Stryker MGS	US/CA	2006
Strv 74	SE	1958
Type 62	CN	1961
Type 63	CN	1963
Type 05 AAV	CN	2005
Type 08 ARV	CN	2008

Open Tanks		
ASU-57	SU	1951
SU-76	SU	1943

Medium IFVs		
BMP-3	SU	1987
Boxer IFV	NL/DE	2011
Strf 9040/CV90	SE	1994
LAV III (add-on-armor)	CA	2009
M2A2/3/4 Bradley	US	1988
M3A2/3/4 Bradley	US	1988
SPz Marder	DE	1971
SPz Puma	DE	2015

Light IFVs		
AIFV	US	1979
AMX-10P	FR	1973
BMP-1	SU	1966
BMP-2	SU	1980
Commando (20 mm)	US	1963
FV432 RARDEN	GB	1975
LAV III	CA	1999
LAV-25	CA	1983
M2 and M2A1 Bradley	US	1981
M3 and M3A1 Bradley	US	1981
M113A1 FSC	AU	1972
Piranha IFV	CH	1974
Pbv 301	SE	1961
Pbv 302	SE	1966
SPz Kurz	FR	1959
SPz Lang HS.30	CH	1960
Stryker ICVD	US	2018
Type 86	CN	1992
Type 92	CN	1992
Type 05 IFV	CN	2005
Type 08 IFV	CN	2008
VCBI	FR	2008
Warrior	GB	1987
YPR-765	NL	1977

Heavy APCs		
Achzarit	IL	1988
Nakpadon	IL	1998
Namer	IL	2008
Puma	IL	1993

Medium APCs		
Boxer APC	NL/DE	2011
Nagmachon	IL	2000
Nagmashot	IL	1990

Light APCs		
Bronco/Warthog	SG	2001
BvS10/Viking	SE	2005
BTR-50PK	SU	1958
BTR-60PA/PAI/PB	SU	1965
BTR-70	SU	1972
BTR-80	SU	1986
BTR-152K	SU	1959
Commando	US	1963
FV432	GB	1963
LAV I Grizzly	CA	1976
LAV II Bison	CA	1990
LVTP-5	US	1956
LVTP-7 (AAV)	US	1972
M59	US	1954
M75	US	1952
M113	US	1960
OT-64 SKOT	CS/PL	1963
OT-810 half-track	CS	1958
Piranha APC	CH	1974
Saracen	GB	1952
Saxon	GB	1983
Spartan	GB	1978
Stryker ICV	US/CA	2002
TPz Fuchs	DE	1979
Type 63	CN	1978
Type 77	CN	1978
Type 89	CN	1978
Type 08 APC	CN	2008
VAB	FR	1979
YP-408	NL	1964

Open APCs		
AT-P	SU	1954
BTR-40	SU	1950
BTR-152A	SU	1950
BTR-50P	SU	1952
BTR-60P	SU	1959
M3 half-track	US	1941
SKP/VKP	SE	1944
SKPF/VKPF	SE	1956

Medium Armored Anti-Tank		
Raketen JPz 1	DE	1961
Raketen JPz 2	DE	1967
Jaguar 1	DE	1978
Jaguar 2	DE	1983

Light Armored Anti-Tank		
BRDM-2 ATGM	SU	1973
BRDM-3 ATGM	SU	1973
Fennek MRAT	DE/NL	2003
FV438 Swingfire	GB	1971
M150	US	1973
M901 ITV	US	1979
Striker	GB	1976
Stryker ATGM	US/CA	2003
VAB HOT	FR	1978
YP-408 PRAT	NL	1976
YPR-765 PRAT	NL	1980

Light Armored Reconnaissance		
BRDM-2	SU	1973
Fennek	DE/NL	2003
Ferret	GB	1952
M113 C&V/Lynx	US	1966
M114	US	1962

Open Armored Reconnaissance		
M20	US	1943

Light Armored Mortar		
FV432 with 81 mm	GB	1963
M106	US	1964
M125	US	1964
M1064	US	1990
Stryker MCV	US/CA	2005

Light Armored Artillery		
2S1 Gvozdika	SU	1972
2S3 Akatsiya	SU	1971
2S19 Msta-S	SU	1989
AMX-30 AuF1	FR	1977
AS-90	GB	1992
Bkan 1	SE	1967
FV433 Abbot	GB	1965
L-33 Ro'Em	IL	1973
M52	US	1955
M108	US	1962
M109	US	1963
Panzerhaubitze 2000	DE	1998

Open Armored Artillery		
AMX-105A	FR	1958
M-50	IL	1962
M7 Priest	US	1942
M40	US	1945
M44	US	1954

Light Armored Rocket Artillery		
M270 MLRS	US	1983
BM-1	RU	1988

Light Truck		
GAZ-69	SU	1953
HMMWV	US	1985
Ilitis	GE	1978
Landrover	UK	1949
M38 1/4-ton truck	US	1949
M151 1/4-ton truck	US	1960
UAZ-469	SU	1971
Willys MB 1/4-ton truck	US	1941

Tracked Carrier		
AT-T/S/L	SU	1950
Bv 202	SE	1964
Bv 206	SE	1980
BvS10/Beowulf	SE	2024
M548	US	1960
Sisu Nasu	FI	1986

Mobile Anti-Tank Missile		
M151 TOW	US	1970
M966 HMMWV TOW	US	1985

Mobile Artillery		
Archer	SE	2016
CAESAR	FR	2004

Tracked Artillery		
2S4 Tyulpan	SU	1975
2S7 Pion	SU	1976
M41	US	1942
M107	US	1962
M110	US	1963
AMX-13 F3	FR	1962

Mobile Rocket Artillery		
BM-8	SU	1941
BM-13	SU	1941
BM-31	SU	1944
BM-24	SU	1947
BM-14	SU	1952
BM-21 Grad	SU	1963
BM-27 Uragan	SU	1975
BM-30 Smerch	SU	1989
M142 HIMARS	US	2005

Tracked Missile Artillery		
M752 Lance	US	1972
AMX-30 Pluton	FR	1974

Mobile Missile Artillery		
9K52 Luna-M (Frog-7)	SU	1962
Corporal	US	1954
Sergeant	US	1962
M34/M50 Honest John	US	1953
M74 Redstone	US	1958
M790 Pershing 1a	US	1962
M1003 Pershing II	US	1983
OTR-21 Tochka (SS-21)	SU	1975
RSD-10 Pioneer (SS-20)	SU	1976

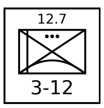
AAA UNITS

AAA units use guns to destroy attacking aircraft, but also have a secondary ground-combat capability. These units are presented below with their counters.

The AAA Unit Table summarizes the properties of AAA units. As well as the basic properties common to all ground units, the table gives: the AAA class (light, medium, or heavy); the maximum altitude; the limits of short, medium, and long range in hexes; the hit rolls at short, medium, and long range; the damage rating; whether the unit has an all-weather or ranging-only fire-control radar and its frequency band; whether it has night infrared sights; and whether it is also equipped with a SAM system. If the unit has a SAM, this will be described in detail in the section on SAM launcher units.[19]

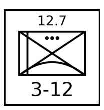
All AAA platoons and batteries are also available as sections (but not squads), with damage resiliences of 2, hit rolls reduced by 1, and VPs corresponding to a platoon or battery that has taken 1D damge.

LIGHT AAA UNITS



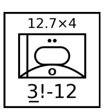
M2 Platoon: The Browning M2 .50 cal (12.7 mm) heavy machine gun is normally found on a low tripod for ground combat, but here is used on a tall dual-purpose mount suitable for air defense. A platoon has six guns.[20]

The M2 gun was introduced in 1933 and has been used by United States forces and their allies in numerous conflicts since then.



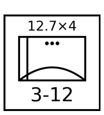
DShK-38 Platoon: The DShK-38 12.7 mm heavy machine gun is most commonly found on a low, wheeled mount for ground combat, but here is used on a tall tripod suitable for air defense. A platoon has six guns.[21]

The DShK-38 entered service with the Soviet Army in 1938 and saw action in World War II and numerous conflicts since then, including the Korean and Vietnam Wars, in which the Soviet Union participated or provided arms.



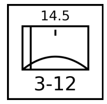
M16 Section: The M16 Multiple Gun Motor Carriage has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns mounted on a modified M3 half-track. A section has two vehicles.[22]

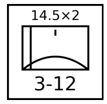
The M16 was used by United States forces during World War II and (primarily as an anti-personnel weapon) in the Korean War for airfield defense. Additionally, Israeli forces employed it during the 1956 Arab-Israeli War.

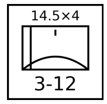


M55 Platoon: The M55 has an M45 turret with four M2 .50 cal (12.7 mm) heavy machine guns mounted on a towed carriage. A platoon has four gun mounts.[23]

The M55 was used by United States forces during the Korean War and by Pakistani forces during the 1965 India-Pakistan War.

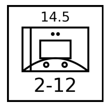


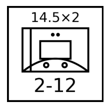


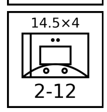


ZPU-1/2/4 Battery: The ZPU-1/2/4 have single, twin, or quad 14.5 mm heavy machine guns in open mounts on towed carriages. A battery has six gun mounts.[24][25][26]

The ZPU-1/2/4 was introduced in 1949 and saw extensive service with Soviet or Soviet-supported forces in many subsequent conflicts, including with communist forces in the Korean War and Vietnam War and with Arab forces in the various Arab-Israeli Wars from 1956 onwards.

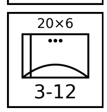






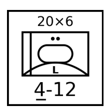
Mobile ZPU-1/2/4 Section: This is a section of two light trucks mounting ZPU-1/2/4 guns.[27][28]

Many users of the ZPU-1/2/4 mount it on a suitable truck for increased mobility. Like the towed versions, it has seen combat in many conflicts.



M167 Platoon: The M167 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in an open mount on a towed carriage. The fire-control system has optical sights and a range-only radar. A platoon has four gun mounts.[29]

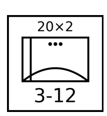
The M167 was introduced into service with the US Army in 1967, principally for airfield defense, and subsequently served with other US allies.



M163 Section: The M163 Vulcan Air Defense System (VADS) has a 20 mm six-barreled Vulcan gun in a lightly armored but open-topped turret on an adapted M113 APC. The fire-control system has optical sights and a range-only radar. A section has two vehicles.[30]

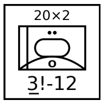
The M163 VADS was introduced into service with the US Army in 1968 and has since been adopted by other US allies. The M163 was used by US forces in the Vietnam War, the 1989 Invasion of Panama, and the 1991 Gulf War. It was also used by the Israel Army in the 1982 Lebanon War and in 2002 during the Second Intifada.

From 1988 in the US Army, each M163 was typically issued with a FIM-92 Stinger launcher with two rounds, so each M163 section effectively carried a mounted FIM-92 section.



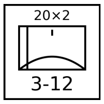
TCM-20 Platoon: The TCM-20 is an Israeli adaptation of the M55, with two Hispano-Suiza 20mm guns from obsolete aircraft replacing the four .50 cal machine guns. As with the M55, the fire-control system is entirely optical. A platoon has four gun mounts.[31]

The TCM-20 was employed by Israeli forces from 1970 and subsequently saw combat in the 1973 Arab-Israeli War and the 1982 Lebanon War.



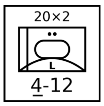
M3 TCM-20 Section: The M3 TCM-20 has a TCM-20 turret mounted in an adapted M3 half-track, like the earlier M16. A section has two vehicles.[32]

The M3 TCM-20 was used by Israeli forces in the 1973 Arab-Israeli War and the 1982 Lebanon War.



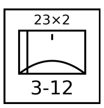
Twin Rh-202 Battery: The Rheinmetall Rh-202 has a single or twin 20mm gun in an open mount on a towed carriage. The fire-control system is entirely optical. The most common version is the twin mount. A battery has six gun mounts.[33]

The twin version entered service with West German forces in 1972 and was used for airfield defense. Subsequently, it was exported and notably used by the Argentine Air Force in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).



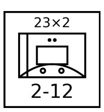
Panhard M3 DCA Section: The Panhard M3 DCA vehicles have an armored turret for twin 20 mm guns mounted on the Panhard M3 APC. The guns have optical sights, but typically one vehicle in each section or platoon is equipped with a search-and-ranging radar and can share this information automatically with the others. A section has two vehicles.[34]

The Panhard M3 DCA served with the armed forces of Côte d'Ivoire, Niger, and the UAE.



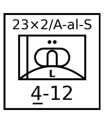
ZU-23 Battery: The ZU-23 has twin 23 mm cannons in an open mount on a towed carriage. It was designed as a more powerful replacement for the ZPU-1/2/4. The fire-control system is entirely optical. A battery has six gun mounts.[35]

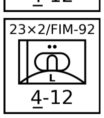
The ZU-23 was introduced into service by the Soviet Army in 1960. It later served with the armed forces of many Soviet allies, including with North Vietnam in the Vietnam War, with Egypt and Syria in the 1967 and 1973 Wars, with Soviet forces in the Soviet-Afghan War, with Iraq in the Iraq-Iran War, with Syria in the 1982 Lebanon War, and with both sides in the Afghan Civil War.



Mobile ZU-23 Section: This is a section of two light trucks mounting ZU-23 guns.[36]

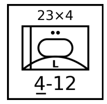
As with the earlier ZPU-1/2/4, many users of the ZU-23 mount it on vehicles, including trucks, tracked carriers, and APCs, for better mobility.





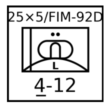
Sinai 23 Section: The Sinai 23 has a Dassault TA20 turret with two 23 mm guns and six Ayn-al-Saqr (similar to the Strela-2M) or Stinger missiles on an M113 chassis. It has night IR sights. A section has two vehicles. Each battery operates with one vehicle equipped with an RA20S search radar.[37]

The Sinai 23 entered service with the Egyptian Army in 1991.



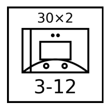
ZSU-23-4 Section: The ZSU-23-4 has four radar-controlled 23 mm guns, similar to those in the ZU-23, mounted in an armored turret on a hull adapted from the PT-76 light tank. A section has two vehicles.[38]

The ZSU-23-4 entered service in the Soviet Army in 1965, replacing the ZSU-57-2 and outclassing all NATO anti-aircraft guns at that time. A platoon of four was assigned to the air-defense battery of tank and motor-rifle regiments and later complemented by a platoon of four SA-9 vehicles. It was exported to Soviet allies and saw extensive combat with Arab forces in the 1973 Arab-Israeli War and with Iraqi forces in the Iran-Iraq War and the 1991 Gulf War.



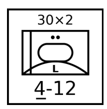
LAV-AD Section: The LAV-AD has a 25 mm GAU-12 rotary cannon and eight FIM-92D Stinger missiles mounted in a turret on the LAV-25 vehicle. It does not have radar, but is equipped with a passive night IR sight. A section has two vehicles.[39]

The LAV-AD entered service with the USMC in 1997, but was retired after only a few years.



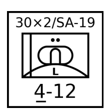
M53/59 Section: The M53/59 has a twin M53 30 mm gun mount on a partially armored truck. The fire-control system is entirely optical. A section has two vehicles.[40]

The M53/59 entered service in the Czech Army in 1959 and was used instead of the ZSU-57-2. It was also exported to Iraq and saw combat in the Iran-Iraq War and the 1991 Gulf War.



AMX-30SA Section: The AMX-30SA is equipped with a pair of radar-controlled 30 mm Hispano-Suiza guns mounted in an armored turret on the hull of an AMX-30 tank. It is an improved version of the AMX-30 DCA. A section has two vehicles.[41]

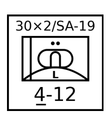
The AMX-30SA served only in the Saudi Army from 1979.



Tunguska Section: The Tunguska has twin radar-controlled 30mm guns and four SA-19 missiles in an armored turret on an armored and tracked hull. It was designed to counter the A-10 and AH-64, against which the earlier ZSU-23-4 had weaknesses. A section has two vehicles.[42]

The Tunguska entered service in the Soviet Army in 1984. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported.

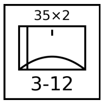
The Tunguska is referred to as the "ZSU-30-2" in *Air Strike*.



Tunguska-M Section: The Tunguska-M is an improved version of the Tunguska with eight missiles instead of four. A section has two vehicles.[43]

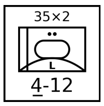
The Tunguska-M entered service in the Soviet Army in 1990. It has been used by Russia, Ukraine, and Belarus after the break-up of the Soviet Union and has been exported.

MEDIUM AAA UNITS



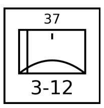
Oerlikon GDF Battery: The Oerlikon GDF has two 35 mm guns in an open mount on a towed carriage. A battery has six gun mounts. The basic fire-control system is optical, but a section is often coupled with the Super Fledermaus or Skyguard fire-control radars. (These radars cannot control a whole battery, only a section.)[44]

The Oerlikon GDF entered service in 1963 and has served widely. Notably, it saw combat in the South Atlantic War with the Argentine forces in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).



Gepard Section: The Gepard has two radar-controlled 35 mm guns in an armored turret on an adapted Leopard 1 tank hull. A section has two vehicles.[45]

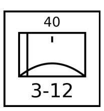
The Gepard entered service in 1976 with the West German Army and also served with the Belgian and Dutch armies. After the end of the Cold War, they were retired by these armies, and many were sold on to other countries.



61-K Battery: The 61-K (M1939) has a single 37 mm gun in an open mount on a towed carriage. The fire-control system is entirely optical, and (as an exception to the normal rule allowing medium AAA units to employ external FCRs) the 61-K cannot be coupled to an external fire-control radar. A battery has six gun mounts.[46]

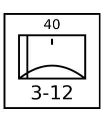
After entering service in 1939 with the Soviet Army, it served in WW2 and after until replaced by the S-60 57 mm gun. It also served with many Soviet allies, and in particular with communist forces in the Korean and Vietnam Wars.

The 61-K is referred to as the "M-38" in *Air Strike*, *Eagles of the Gulf*, and *The Speed of Heat*.[47]



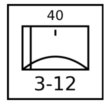
Bofors L/60 Battery: The Bofors L/60 has a single 40 mm gun in an open mount on a towed carriage. The fire-control system is entirely optical. A battery has six gun mounts.[48]

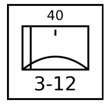
It entered service in the 1940s and was widely used by Allied forces during WW2, both on land and sea. After the war, many continued to see service, although it was increasingly inadequate against faster jet aircraft and in many cases was replaced by the new L/70 model.



Bofors L/70 Battery: The Bofors L/70 has a single 40 mm gun in an open mount on a towed carriage. Between the 1930s and the end of WW2, aircraft speed increased substantially, and so the engagement time of the Bofors L/60 became inadequate. The new L/70 fired a lighter shell at a higher velocity and achieved a significant improvement in range and engagement time. The basic fire-control system is optical, but various external fire-control radars can be used with it, including the Super Fledermaus and Flycatcher systems. A battery has six gun mounts.[49]

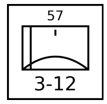
NATO forces and others adopted it widely starting in 1952.





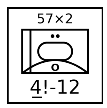
Bofors L/70 BOFI and BOFI-R Battery: The BOFI version of the Bofors L/70 adds modern sights, a laser rangefinder, and proximity-fuzed shells. The BOFI-R version also adds an integrated fire-control radar. A battery has six gun mounts.[50]

The BOFI is used by the armed forces of Brazil.[51]



S-60 Battery: The S-60 is a Soviet single 57 mm gun on an open mount on a towed carriage. The basic fire-control system is optical, but the gun is often used with the SON-9 (Fire Can) radar. A battery has six gun mounts.[52]

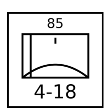
It entered service in 1950 with Soviet forces as a replacement for the 61-K 37 mm gun. It was exported to many Soviet allies and saw combat with Arab forces in the 1967 and 1973 Arab-Israeli War, with communist forces in the Vietnam War, and with Iraqi forces in the Iran-Iraq War and the Gulf War. However, it was not used by communist forces in the Korean War. When paired with the SON-9 (Fire Can) radar, it was considered one of the more dangerous AAA systems faced by US aircraft in Vietnam.



ZSU-57-2 Section: The ZSU-57-2 has two 57 mm cannons in a lightly armored but open-topped turret on an armored and tracked hull. The cannons are similar to the S-60 cannons. The fire-control system is optical. A section has two vehicles.[53]

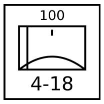
The ZSU-57-2 entered service in the Soviet Army in 1955 and replaced the BTR-40A and BTR-152A self-propelled 14.5 mm AA guns in the AA batteries of tank regiments. It was in turn replaced by the ZSU-23-4 from 1965. The ZSU-57-2 also served widely with Soviet allies and saw combat with Egypt and Syria in the 1967 and 1973 Wars, with Syria in the 1982 Lebanon War, with North Vietnam in the 1972 Easter Offensive and the 1975 Spring Offensive, and with various factions in the Yugoslav Wars in the 1990s.

HEAVY AAA UNITS



KS-12 Battery: The KS-12 (M1939 52-K) and the very similar KS-12A (M1944 KS-1) have a single 85 mm gun in an open mount on a towed carriage. The fire-control system is optical, but the guns are often paired with the SON-9 (Fire Can) radar. A battery has six gun mounts.[54]

It entered service with Soviet forces in 1939. After WW2, it began to be replaced in Soviet service by the KS-19 100 mm and KS-30 130 mm guns, but was widely used by Soviet allies and saw combat with communist forces in the Korean War and the Vietnam War.



KS-19 Battery: The KS-19 has a single 100 mm gun in an open mount on a towed carriage. The fire-control system is optical, but like other contemporary Soviet guns, it was often paired with the SON-9 (Fire Can) radar. A battery has six gun mounts.[55]

It entered service in 1948 with Soviet forces as a replacement for the KS-12 85 mm gun. It also served with many Soviet allies and saw combat with communist forces in the Korean War and Vietnam War and Iraqi forces in the Iran-Iraq and Gulf Wars.

AAA Units Table

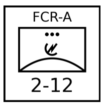
Type	Size[a]	Year	Defense Strength	Sighting Range	Mobility	VPs	Gun	AAA Class	Altitude	Range			Hit			Damage Rating	FCR	Night IR Sights	SAM
						3D/2D/D				S	M	L	S	M	L				
M2	Platoon	1933	3	12	U	3/2/1	12.7 mm	L	5	2	3	4	3	2	1	1	—	—	—
DShK-38	Platoon	1938	3	12	U	3/2/1	12.7 mm	L	5	2	3	4	3	2	1	1	—	—	—
M16	Section	1944	3!	12	U	6/3	12.7 mm × 4	L	5	2	3	4	4	3	2	2	—	—	—
M55	Platoon	1945	3	12	T	4/3/1	12.7 mm × 4	L	5	2	3	4	5	4	3	2	—	—	—
ZPU-1	Battery	1949	3	12	T	3/2/1	14.5 mm	L	6	2	3	5	3	2	1	1	—	—	—
ZPU-2	Battery	1949	3	12	T	4/3/1	14.5 mm × 2	L	6	2	3	5	4	3	2	1	—	—	—
ZPU-4	Battery	1949	3	12	T	5/3/2	14.5 mm × 4	L	6	2	3	5	5	4	3	2	—	—	—
Mobile ZPU-1	Section	1949	2	12	G/P	2/1	14.5 mm	L	6	2	3	5	2	1	0	1	—	—	—
Mobile ZPU-2	Section	1949	2	12	G/P	3/2	14.5 mm × 2	L	6	2	3	5	3	2	1	1	—	—	—
Mobile ZPU-4	Section	1949	2	12	G/P	4/2	14.5 mm × 4	L	6	2	3	5	4	3	2	2	—	—	—
M167	Platoon	1967	3	12	T	7/5/2	20 mm × 6	L	8	2	4	6	6	5	4	3	R/HF	—	—
M163	Section	1968	4	12	U	8/4	20 mm × 6	L	8	2	4	6	5	4	3	3	R/HF	—	—
TCM-20	Platoon	1970	3	12	T	5/3/2	20 mm × 2	L	8	2	4	6	4	4	3	2	—	—	—
M3 TCM-20	Section	1970	3!	12	U	6/3	20 mm × 2	L	8	2	4	6	3	3	2	2	—	—	—
Single Rh-202	Battery	1972	3	12	T	4/3/1	20 mm	L	8	2	4	6	4	3	2	2	—	—	—
Twin Rh-202	Battery	1972	3	12	T	5/3/2	20 mm × 2	L	8	2	4	6	5	4	3	3	—	—	—
Panhard M3 DCA	Section	1975	4	12	U	7/4	20 mm × 2	L	8	2	4	6	4	4	3	2	R/UF	—	—
ZU-23	Battery	1960	3	12	T	6/4/2	23 mm × 2	L	9	2	5	8	5	4	3	3	—	—	—
Mobile ZU-23	Section	1960	2	12	G/P	5/3	23 mm × 2	L	9	2	5	8	4	3	2	3	—	—	—
Sinai 23 Ayn-al-Saqr	Section	1992	4	12	U	10/5	23 mm × 2	L	9	2	5	8	4	3	2	3	—	Y	Ayn-al-Saqr
Sinai 23 Stinger	Section	1992	4	12	U	12/6	23 mm × 2	L	9	2	5	8	4	3	2	3	—	Y	FIM-92
ZSU-23-4	Section	1965	4	12	U	9/5	23 mm × 4	L	9	2	5	8	5	4	3	3	W/VF	—	—
LAV-AD	Section	1997	4	12	U	14/7	25 mm × 5	L	10	3	6	9	4	3	2	4	—	Y	FIM-92D
M53/59	Section	1959	3	12	G/P	7/4	30 mm × 2	L	11	4	7	9	3	2	1	3	—	—	—
AMX-30SA	Section	1979	4	12	U	10/5	30 mm × 2	L	11	4	7	9	5	4	3	3	W/VF	—	—
Tunguska	Section	1984	4	12	U	20/10	30 mm × 2	L	11	4	7	9	5	4	3	4	W/VF	—	SA-19
Tunguska-M	Section	1990	4	12	U	22/11	30 mm × 2	L	11	4	7	9	5	4	3	4	W/VF	—	SA-19
Oerlikon GDF	Battery	1963	3	12	T	7/5/2	35 mm × 2	M	12	4	8	10	4	3	2	4	—	—	—
Oerlikon GDF	Section	1963	3	12	T	5/2	35 mm × 2	M	12	4	8	10	3	2	1	4	—	—	—
Gepard	Section	1976	4	12	U	11/6	35 mm × 2	M	12	4	8	10	5	4	3	4	W/VF	—	—
61-K	Battery	1939	3	12	T	5/3/2	37 mm	M	12	4	8	11	3	2	1	4	—	—	—
Bofors L/60	Battery	1935	3	12	T	5/3/2	40 mm	M	15	4	8	12	2	2	1	4	—	—	—
Bofors L/70	Battery	1952	3	12	T	6/4/2	40 mm	M	15	4	8	12	3	3	2	4	—	—	—
Bofors L/70 BOFI	Battery		3	12	T	8/5/3	40 mm	M	15	4	8	12	5	5	4	4	—	—	—
Bofors L/70 BOFI-R	Battery		3	12	T	9/6/3	40 mm	M	15	4	8	12	6	6	5	4	W/VF	—	—
S-60	Battery	1950	3	12	T	8/5/3	57 mm	M	18	5	10	15	2	2	1	5	—	—	—
ZSU-57-2	Section	1955	4!	12	U	6/3	57 mm × 2	M	18	5	10	15	2	1	1	5	—	—	—
KS-12	Battery	1939	4	18	T	9/6/3	85 mm	H	27	6	12	18	2	1	0	6	—	—	—
KS-19	Battery	1948	4	18	T	10/7/3	100 mm	H	39	7	14	21	1	1	0	6	—	—	—

[a] Platoons and batteries are available as sections with corresponding modifications to their damage resiliences and hit rolls.

FCR UNITS

Most medium and heavy AAA batteries can be associated with an add-on fire-control radar (FCR), which improves their accuracy and allows them to engage unsighted targets at night and in adverse weather. These units are presented below with their counters.

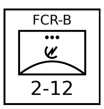
The FCR Unit Table summarizes the properties of FCR units. As well as the basic properties common to all ground units, the table gives: the FCR class; the frequency band; and the modifier to the hit roll.



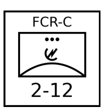
Towed FCR-A Platoon: FCR-As typically use 1940s technology. They include the SON-9 FCR.

SON-9: The Soviet SON-9 (Fire Can) FCR-A was derived from the earlier SON-4, which in turn was derived from the US-supplied SCR-584. It is commonly used with 57 mm S-60, 85 mm KS-12, 100 mm KS-19, and 130 mm KS-30 batteries.

The SON-9 entered service in 1955 with Soviet forces and was widely exported to Soviet allies. It has seen combat, including with North Vietnam in the Vietnam War, Egypt and Syria in the 1967 War, the War of Attrition, and the 1973 War.[56]



Towed FCR-B Platoon: FCR-Bs typically use 1950s technology.



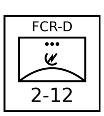
Towed FCR-C Platoon: FCR-Cs typically use 1960s technology. They include the Super Fledermaus and Flycatcher FCRs.

Super Fledermaus: The Contraves Super Fledermaus FCR-C can control a section of two Oerlikon GDF 35 mm guns; it cannot control a whole battery.

It entered service in 1965 with the Swiss Air Force and also was used by the Argentine Air Force, Brazil, Iran, Japan, and South Africa. It saw combat in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport).[57]

Flycatcher: The Signaal Flycatcher FCR-C can control up to six Bofors L/60 or L/70 guns or SAM launchers.

Flycatcher entered service with the Royal Netherlands Air Force in 1979. It subsequently served with the Indian Armed Forces from 1985, the Irish Defense Forces from 2003, the Royal Netherlands Army from 1987, the Thai Armed Forces from 1981, Turkey, and Venezuela.



Towed FCR-D Platoon: FCR-Ds typically use 1970s technology. The include the Skyguard.

FCR Unit Table

Type	Size	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	FCR Class	FCR Frequency	FCR Modifier
Towed FCR-A	Platoon	2	12	T	6/4/2	A	LF	−1
Towed FCR-B	Platoon	2	12	T	6/4/2	B	MF	−2
Towed FCR-C	Platoon	2	12	T	7/5/2	C	VF	−2
Towed FCR-D	Platoon	2	12	T	8/5/3	D	MW	−3

Skyguard: The Contraves Skyguard FCR-D can control two Oerlikon GDF 35 mm guns and additionally a SAM launcher with AIM-9 Sparrow, RIM-9 Sea Sparrow, or Aspide missiles.

Skyguard entered service in 1977 with the Swiss Air Force. It was subsequently very widely exported and used by the Argentine Army, Austria, Bahrain, Bangladesh, Brazil, Canada, Chile, China, Cyprus, Egypt, Greece, Indonesia, Iran, Italy, South Korea, Kuwait, Malaysia, Oman, Pakistan, Saudi Arabia, Spain, Taiwan, and the UK. It saw combat with the Argentine Army in the 1982 South Atlantic War in the defense of BAM Malvinas (Port Stanley Airport) and BAM Condor (Goose Green Airfield).

SAM UNITS

SAM launcher units are specialized in the destruction of attacking aircraft with surface-to-air missiles.

Properties: Table **??** summarizes the properties of infantry SAM units, and Figure **??** shows their representation as counters. In addition to the properties shared by all ground units, Table **??** gives:

– the number of ready, volley, and reload missiles;
– the multi-target capability;
– whether the unit has quick-reaction capability;
– whether the unit has IFF capability;
– whether the unit has night IR sights; and
– the optical lock-on number and range.

Missile Properties: Table **??** summarizes the properties of the actual missiles:

– guidance mode;
– launch roll;
– turn rate;
– flight time;
– visibility;
– ECCM, chaff, and flare ratings;
– whether the missile is active homing (AH) or has a home-on-jam (HOJ) capability;
– boost phase in FPs;
– base speed and sustainer duration in game turns;
– minimum altitude (and modifier for targets at T level);
– hit rolls for direct and proximity hits; and
– damage ratings for direct and proximity hits.

Infantry SAM Units: Infantry SAM units are squads of infantry specialized in the destruction of attacking aircraft with portable SAMs. Each squad has one launcher and, unless otherwise specified, one ready missile and two reload missiles.

Infantry SAM squads may be attached to infantry, infantry weapons, or infantry HQ platoons.

Most infantry SAMs do not have true proximity fuzes. Thus, proximity hits in the game correspond to glancing direct hits in reality.

SA-7A/B Squad: The 9K32 Strela-2 (SA-7A Grail) was the first Soviet portable SAM. It has a rear-aspect, uncooled IR seeker.

The 9K32M Strela-2M (SA-7B Grail) is similar, but uses a rear-aspect, electrically-cooled IR seeker. In Warsaw Pact armies, the Strela-2M was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions.[58][59]

The Strela-2 and -2M entered service with Soviet forces in 1968 and 1970. They have seen combat in almost every conflict since 1970 in which the Soviet Union or its successor states have been participants or supporters. In particular, the Strela-2M was used heavily by North Vietnamese forces in the 1973 Easter Offensive, by Arab forces in the 1973 Yom Kippur War, by Angolan forces in the South African Border War, by the Afghan Mujahedeen in the Soviet-Afghan War (having been supplied by the CIA), by Iran in the Iran-Iraq War in the 1980s, and by Serbian and Serbian-supported forces in the Yugoslav Wars in the 1990s.

SA-14 Squad: The 9K34 Strela-3 (SA-14 Gremlin) is a development of the earlier Strela-2M. It has a rear-aspect, cryogen-cooled IR seeker. In Warsaw Pact armies, the Strela-3 was sometimes used with the PRP portable RWR receiver that was capable of passively identifying enemy aircraft by their radar emissions.[60]

The Strela-3 entered service with the Soviet Army in 1974. It has seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. In particular, it was used by Iraqi forces in the 1991 Gulf War.

SA-16/18 Squad: 9K310 Igla-1 (SA-16 Gimlet) and 9K38 Igla (SA-18 Grouse) use very similar missiles and launchers, but the Igla-1 has a single-channel IR seeker similar to the SA-14, and the Igla uses a dual-channel IR seeker to reduce its susceptibility to flares. The warhead was similar to the SA-7/14, but also exploded any remaining rocket fuel. In the game, this is simulated by increasing the damage rating by one if the missile attacks before expending more than half of its FPs (round down). Both incorporate IFF interrogators, but these were only used in Warsaw Pact armies.[61]

The Igla-1 entered service with the Soviet Army in 1981 and the Igla in 1983. Both have been widely exported and have seen combat in many conflicts since then in which the Soviet Union or its successor states have been participants or supporters. The Igla was used by Iraqi forces in the 1991 Gulf War, by the Peruvian Army in the 1995 Cenepa War, and by Serbian forces in the 1999 Kosovo War.

FIM-43C Squad: The FIM-43C Redeye Block III was the first US infantry SAM to reach production and enter service. It has a rear-aspect, cryogen-cooled IR seeker.[62]

The FIM-43C entered service with the US Army in 1968 and subsequently was used by Australia, Chad, Denmark, West Germany, Greece, Israel, Jordan, Saudi Arabia, Somalia, Sudan, Sweden, and Turkey. It was also supplied by the CIA to the Afghan Mujahedeen and the Nicaraguan Contras.

FIM-92A/B/C/D Squad: The FIM-92A Stinger has an all-angle, second-generation, cryogen-cooled IR seeker. Its warhead weight 3 kg and was significantly more lethal than in earlier missiles. The FIM-92B Stinger POST adds a UV seeker to distinguish flares from aircraft. The FIM-92C/D Stinger RMP have software updates to further improve the performance against flares. All have an IFF interrogator to help avoid fratricide.[63]

The FIM-92A entered service with the US Army in 1981, the FIM-92B in 1986, the FIM-92C in 1989, and the FIM-92D in 1992. They were exported to dozens of US allies including the Afghan Mujahedeen and the Angolan UNITA. They saw service in the 1982 South Atlantic War, the Soviet-Afghan War, the Angolan Civil War, the Libyan Invasion of Chad, the Tajik Civil War, the Chechen War, the Sri Lankan Civil War, the Syrian Civil War, and the Russian Invasion of Ukraine.

Blowpipe Squad: Blowpipe is an MCLOS optically-guided missile. While in theory this gave it an all-aspect capability, in practice guiding the missile to the target in combat proved to be very difficult.[64]

Blowpipe entered service with the British Army and Royal Marines in 1975. It was also used by the Argentine Army, Canadian Army, Chilean Army and Navy, Ecuadorean Army, Guatemalan Army, Portuguese Army, Malawian Army, Malaysian Army, Nigerian Army, Omani Army, Royal Thai Air Force and Army, and the UAE Army. It saw combat in the 1982 South Atlantic War with both sides and had a low success rate. In 1986, it was trialed by the Afghan Mujahedeen, but again was not considered to be effective. Finally, it was also used in the 1995 Cenepa War by Ecuador.

Javelin/Starburst Squad: Given the poor performance of Blowpipe in the 1982 South Atlantic War, Javelin was quickly developed as an upgraded version of Blowpipe with SACLOS guidance in place of MCLOS guidance. Starburst then replaced the radio-link with laser guidance. Javelin is also called “Javelin GL” and Starburst was originally called “Javelin S15”.[65]

Javelin replaced Blowpipe in the British Army and Royal Marines in 1984. It was also used by the Botswana Army, the Canadian Army, the Malaysian Army, the Peruvian Army, and the South Korean Army.

Starburst in turn replaced Javelin in the British Army and Royal Marines in 1989. It was also used by the Canadian Army, the Malaysian Army, the Qatari Army, and the Royal Thai Army. Starburst served with the British Army in the 1991 Gulf War, but was not used in combat.

Starstreak Squad: Starstreak is a high-speed laser-guided missile. As the missile approaches the target, it launches three independent explosive darts. The darts have contact fuzes, but not proximity fuzes.[66]

Starstreak replaced Javelin in the British Army and Royal Marines in 2000 and is also used by the South African Army and the Armed Forces of Ukraine. Starstreak served with the British Army in the 2003 Invasion of Iraq, but was not used in combat. It has seen combat in the Russian Invasion of Ukraine.

Starstreak LML Squad: The Starstreak Light Multiple Launcher (LML) has three ready-to-fire Starstreak missiles on a pedestal mount with a single sighting unit. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit.[67]

Starstreak LML entered service with the British Army and Royal Marines in 2000. It is also used by Indonesia, Malaysia, and South Africa.

Mistral Squad: The Mistral is a French IR-homing missile on a pedestal mount. The missile has a considerably larger warhead than either the Stinger or Igla missiles, but this comes at a cost in portability. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit.[68]

The Mistral entered service with the French Army in 1990. It was subsequently exported to many other countries.

RBS 70 Squad: The RBS 70 is a Swedish laser-guided missile on a pedestal mount. The launcher and its associated equipment are heavy and will normally be transported by a vehicle unit.[69]

The RBS 70 entered service with the Swedish Army in 1977. It was subsequently used by Argentina, Australia, Bahrain, Brazil, the Czech Republic, Iran, Ireland, Lithuania, Norway, Pakistan, Singapore, Thailand, Tunisia, the UAE, Ukraine, and Venezuela.

Ayn-al-Saqr Squad: The Ayn-al-Saqr (“Eye of the Hawk”) missile is a licensed version of the 9M32M Strela-2m (SA-7B Grail) manufactured in Egypt. It equips one version of the Sinai 23 air-defense vehicle.[70]

It entered service with the Egyptian Army in 1984.

Vehicular and Towed SAM Units:

Echelon: The size of a SAM launcher unit is squad, section, or platoon according to its damage resilience (1D, 2D, or 3D, respectively). This does not always coincide with the actual designations used historically. For example, the SA-2 battery in the game corresponds to a battalion in reality.

Nuclear Warheads: The SA-2E has the option of a nuclear warhead.

Soviet and Russian SAM Launcher Units:

SA-2B/C/E/F Battery: This is an S-75 (SA-2 Guideline) battery of six single launchers for the long-range V-750 missile and their associated P-12 (Spoon Rest) EWR and SNR-75 (Fan Song) TTR. The variants are the S-75 Desna (SA-2B), S-75M Volkhov (SA-2C), S-75AK (SA-2E), and S-75SM (SA-2F). All are radar-guided, but the SA-2F also has an optical backup.

The S-75 Dvina (SA-2A) system was deployed by the Soviet PVO in 1957. In addition to serving as a key part of the air-defense system of the Soviet Union (and its notable role in ending reconnaissance overflights by shooting down the U-2 of Gary Powers) and its allies, it saw combat in Cuba in 1962, with North Vietnamese forces from 1965, with Indian forces in the 1965 war with Pakistan, and with Egyptian and Syrian forces from 1967.

SA-3B Battery: This is an S-125 Neva (SA-3B Goa) battery with four quad launchers for the 5V24 missile and their associated P-15 (Flat Face) EWR and SNR-125 (Low Blow) TTR.

It saw combat with Egyptian and Syrian forces from 1970.

Mobile SA-3B Battery: This is a mobile S-125 Neva (SA-3B Goa) battery with four twin launchers for the 5V24 missile on trucks and their associated P-15 (Flat Face) EWR and SNR-125 (Low Blow) TTR.

It saw combat with Egyptian and Syrian forces from 1970 and with Iraqi forces in the Iran-Iraq War, the Gulf War, and the Invasion of Iraq.

SA-4A/B Battery: This is an armored 2K11 Krug (SA-4A/B Ganef) battery with three launcher vehicles each with two long-range 9M8M1/9M8M2 missiles and their associated P-40 (Long Track) EWR and 1S32 (Pat Hand) TTR.

It saw combat with Egyptian and Syrian forces from 1970 and with Iraqi forces in the Iran-Iraq War, the Gulf War, and the Invasion of Iraq.

SA-6A/B Battery: This is an armored 2K12 Kub (SA-6A/B Gainful) battery with four launcher vehicles, each with three medium-range 2K12 Kub (SA-6A) or 3M9M3 (SA-6B) missiles and their associated P-40 (Long Track) EWR and 1S32 (Pat Hand) TTR.

It saw combat with Egyptian and Syrian forces from 1970 and

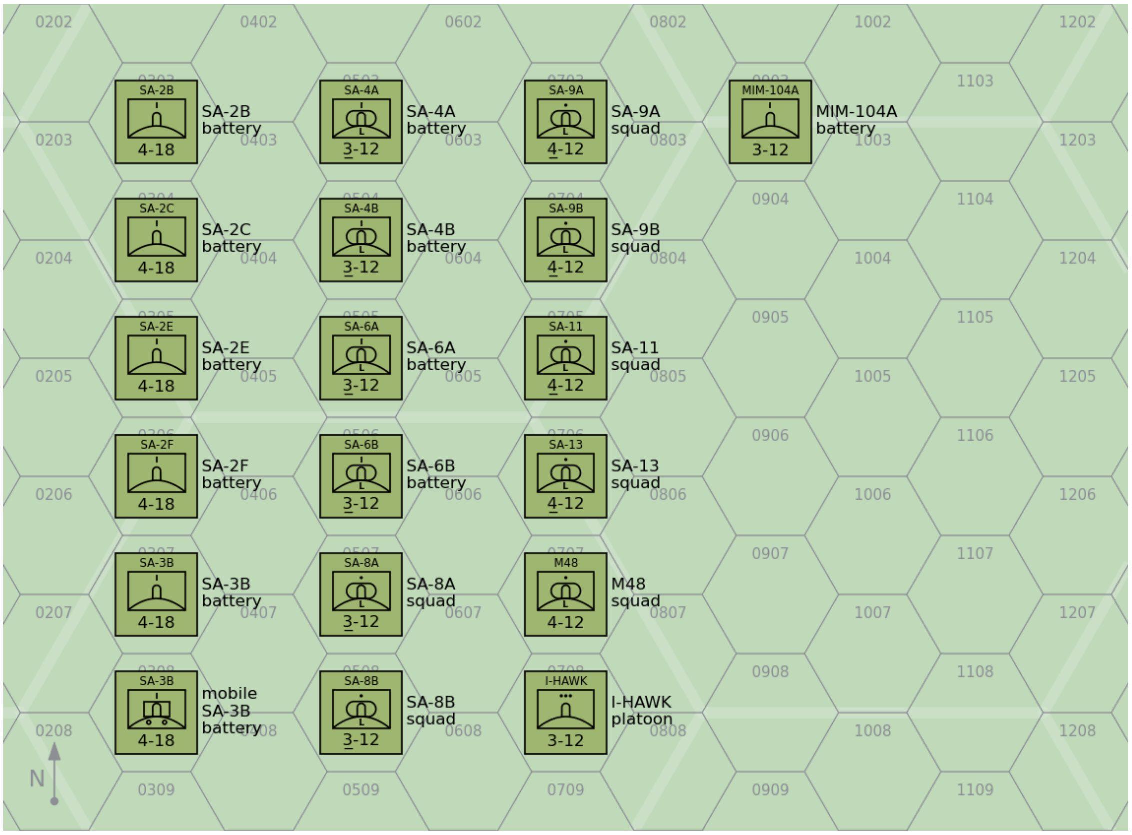


Figure 3.1: SAM Launcher Units

with Iraqi forces in the Iran-Iraq War, the Gulf War, and the Invasion of Iraq.

SA-8A/B Squad: This is a squad with a single 9K33 Osa or 9K33M Osa-AK (SA-8A/B Gecko) vehicle with integrated 1S51M3 (Land Roll) TTR. The SA-8A carries four 9M33 missiles, and the SA-8B carries six 9M33M missiles.

The Osa has been widely exported and seen combat with Syrian forces in the 1982 Lebanon War, with Cuban forces in the South African Border War, with Iraqi forces in the 1990 Gulf War, and in more recent conflicts.

SA-9A/B Squad: This is a squad with a single 9K21 Strela-1 or 9K21M Strela-1M (SA-9A/B Gaskin) vehicle with four 9M21 or 9M21M missiles. In the Soviet army, a platoon of four was assigned to the air-defense battery of tank and motor-rifle regiments, complementing the ZSU-23-4.

The Strela-1 has been widely exported and saw combat in the South African Border War, the Iran-Iraq War, the Lebanon War, the Gulf War, the Kosovo War, and the Invasion of Iraq.

SA-11 Squad: This is a squad with a single 9K37 Buk (SA-11 Gadfly) vehicle with four 9M37 missiles and an integrated 9S35 (Fire Dome) TTR. Three squads make up a battery, and three batteries make up a battalion. Normally, one armored 9S18 (Tube Arm) or 9S18M1 (Snow Drift) EWR is allocated to each battalion.

The SA-11 has seen combat in the War in Abkhazia, the Russo-Ukrainian War, and during the Syrian Civil War.

SA-13 Squad: This is a single 9K35 Strela-10 (SA-13 Gopher) vehicle with four infra-red homing 9M37 missiles. Four squads form a platoon. It began to replace the SA-9B in the air-defense platoons of tank and motor-rifle regiments in the Soviet Army starting in 1976 and was subsequently widely exported to Soviet allies. Compared to the earlier system, the missile offers greater lethality, and the vehicle greater protection and mobility (although it is not amphibious). Some are equipped with 9S16 (Flat Box-B) RWR.

The SA-13 saw combat in the South African Border War with Angolan forces, the Gulf War with Iraqi forces, and in the Kosovo War with Serbian forces.

US SAM Launcher Units:

M48 Squad: This is a single M48 Chaparral vehicle with four adapted Sidewinder infrared-homing missiles. It was developed after the ambitious Mauler SAM program was abandoned and introduced in 1969. Night IR sights were added in 1984 to give it a limited night capability. The MIM-72A was the original missile based on the AIM-9D. The MIM-72C added an improved seeker and warhead, and the MIM-72E/F added a smokeless motor. The final MIM-72G added a seeker based on that of the FIM-92B Stinger, with much improved rejection of infrared countermeasures.

I-HAWK Platoon: This consists of three triple launchers for the MIM-23A/C/D missile, plus supporting radars and command elements. The MIM-23B Improved HAWK and its associated improved radars were developed from the original HAWK MIM-23A to give much improved performance at low levels. The MIM-23C has improved ECCM, and the MIM-23D adds a home-on-jam capability. Two platoons with an EWR make up a battery.

The I-HAWK was deployed by the US Army, US Marines, and many US allies.

MIM-104 Battery: An MIM-104 Patriot battery consists of six launchers with associated radar and command elements. The MIM-104A uses the “Standard” missile and was capable only of defense against aircraft. Patriot is used by the US Army and numerous US allies.

Figure 3.2: Air-Defense Units: Infantry SAM Units

Table 3.1: Air-Defense Units: Infantry SAM Units

						VPs		Missiles							Lock-On		
Name	Size	Year	Defense Strength	Sighting Range	Mobility	3D/2D/D	SAM	Ready	Volley	Reload	Multi-Target	Quick Reaction	IFF	Night IR Sights	Optical	Range	Other Names
SA-7A	Squad	1968	5	6	U	–/–/3	SA-7A	1	1	2	—	Y	—	—	7	9	9K32 Strela-2 (Grail)
SA-7B	Squad	1972	5	6	U	–/–/4	SA-7B	1	1	2	—	Y	—	—	7	9	9K32M Strela-2M (Grail)
SA-14	Squad	1974	5	6	U	–/–/6	SA-14	1	1	2	—	Y	—	—	7	12	9K34 Strela-3 (Gremlin)
SA-16	Squad	1981	5	6	U	–/–/8	SA-16	1	1	2	—	Y	Y	—	7	12	9K310 Igla-1 (Gimlet)
SA-18	Squad	1983	5	6	U	–/–/10	SA-18	1	1	2	—	Y	Y	—	8	16	9K38 Igla (Grouse)
FIM-43C	Squad	1968	5	6	U	–/–/4	FIM-43C	1	1	2	—	Y	—	—	7	9	Redeye Block III
FIM-92A	Squad	1981	5	6	U	–/–/9	FIM-92A	1	1	2	—	Y	Y	—	8	12	Stinger
FIM-92B	Squad	1986	5	6	U	–/–/10	FIM-92B	1	1	2	—	Y	Y	—	8	12	Stinger POST
FIM-92C	Squad	1989	5	6	U	–/–/10	FIM-92C	1	1	2	—	Y	Y	—	8	12	Stinger RMP
FIM-92D	Squad	1992	5	6	U	–/–/10	FIM-92D	1	1	2	—	Y	Y	—	8	12	Stinger RMP
Blowpipe	Squad	1975	5	6	U	–/–/6	Blowpipe	1	1	2	—	Y	Y	—	7	9	
Javelin	Squad	1984	5	6	U	–/–/8	Javelin	1	1	2	—	Y	Y	—	7	9	Javelin GL
Starburst	Squad	1989	5	6	U	–/–/8	Starburst	1	1	2	—	Y	Y	—	7	9	Javelin S15
Starstreak	Squad	2000	5	6	U	–/–/12	Starstreak	1	1	2	—	Y	Y	—	7	12	
Starstreak LML	Squad	2000	4	6	U	–/–/14	Starstreak	3	1	0	—	Y	Y	—	7	12	
Mistral	Squad	1990	4	6	U	–/–/11	Mistral	1	1	2	—	Y	—	—	7	12	
RBS 70	Squad	1977	4	6	U	–/–/8	RBS 70	1	1	2	—	Y	Y	—	7	10	
Ayn-al-Saqr	Squad	1984	5	6	U	–/–/4	Ayn-al-Saqr	1	1	2	—	Y	—	—	7	12	

Table 3.2: Air-Defense Units: SAM Launcher Units

Name	Size	Year	Defense Strength	Sighting Range	Mobility	VPs 3D/2D/D	SAM	EWR Frequency	EWR Range	EWR MTI	TTR Frequency	TTR Range	Missiles Ready	Missiles Volley	Missiles Reload	Multi-Target	Quick Reaction	Night IR Sights	Lock-On Radar	Lock-On Optical	Lock-On Range	Other Names
SA-2B	Battery	1957	4	18	G/P	8/5/3	SA-2B	LF	30	—	LF	20	6	2	0	—	—	—	6	—	—	S-75 Desna (Guideline)
SA-2C	Battery	1960	4	18	G/P	8/5/3	SA-2C	LF	30	—	MF	20	6	2	0	—	—	—	6	—	—	S-75M Volkhov (Guideline)
SA-2E	Battery	1962	4	18	G/P	8/5/3	SA-2E	LF	30	—	MF	20	6	2	0	—	—	—	6	—	—	S-75AK (Guideline)
SA-2F	Battery	1967	4	18	G/P	8/5/3	SA-2F	LF	30	—	LF	20	6	2	0	—	—	—	6	6	—	S-75SM (Guideline)
SA-3B	Battery	1964	4	18	G/P	10/7/3	SA-3B	MF	35	—	VF	10	16	2	0	—	—	—	7	6	—	S-125M Neva-M (Goa)
Mobile SA-3B	Battery	1964	4	18	G/P	12/8/4	SA-3B	MF	35	—	VF	10	16	2	0	—	—	—	7	6	—	S-125M Neva-M (Goa)
SA-4A	Battery	1963	3	12	U	10/7/3	SA-4A	VF	50	—	HF	30	6	2	0	—	—	—	7	—	—	2K11 Krug (Ganef)
SA-4B	Battery	1973	3	12	U	10/7/3	SA-4B	VF	50	—	HF	30	6	2	0	—	—	—	7	—	—	2K11M Krug-M (Ganef)
SA-6A	Battery	1966	3	12	U	11/7/4	SA-6A	MF	24	—	HF	18	12	2	0	—	—	—	7	6	—	2K12 Kub (Gainful)
SA-6B	Battery	1977	3	12	U	12/8/4	SA-6B	MF	24	—	HF	18	12	2	0	—	—	—	7	6	—	2K12M3 Kub-M3 (Gainful)
SA-8A	Squad	1967	3	12	U	–/–/10	SA-8A	HF	10	—	VF	6	4	2	0	2	Y	—	8	—	—	9K33 Osa (Gecko)
SA-8B	Squad	1980	3	12	U	–/–/10	SA-8B	HF	10	—	VF	6	6	2	0	2	Y	—	8	7	—	9K33M2 Osa-AK (Gecko)
SA-9A	Squad	1968	4	12	U	–/–/7	SA-9A	—	—	—	—	—	4	2	0	2	Y	—	—	7	12	9K31 Strela-1 (Gaskin)
SA-9B	Squad	1970	4	12	U	–/–/7	SA-9B	—	—	—	—	—	4	2	0	2	Y	—	—	7	12	9K31M Strela-1M (Gaskin)
SA-11	Squad	1980	4	12	U	–/–/12	SA-11	—	—	—	HF	18	4	2	0	2	—	—	8	7	—	9K37 Buk (Gadfly)
SA-13	Squad	1976	4	12	U	–/–/10	SA-13	—	—	—	—	—	4	2	0	2	Y	—	—	8	12	9K35 Strela-10 (Gopher)
M48	Squad	1969	4	12	U	–/–/9	MIM-72	—	—	—	—	—	4	2	0	2	Y	Y	—	7	12	Chaparral
I-Hawk	Platoon	1971	3	12	G/P	12/8/4	MIM-23B	—	—	—	VF	20	9	2	0	—	Y	—	7	7	—	
MIM-104A	Battery	1984	3	12	G/P	45/30/15	MIM-104A	LF	60	Y	LF	40	24	8	0	8	Y	—	9	—	—	Patriot (Standard)

Table 3.3: SAMs

Name	Year	Guidance Mode	Launch Roll	Turn Rate	Flight Time	Visibility	ECCM	Chaff	Flare	AH	HOJ	Instant Arming	Boost Phase	Base Speed	Sustainer	Minimum Altitude	Hit Direct	Hit Proximity	Damage Direct	Damage Proximity	Other Names
SA-7A	1968	I	6	BT	1	6	—	—	5	—	—	Y	—	10	0	T	5	6	4	2	9M32 Strela-2 (Grail)
SA-7B	1972	I	7	BT	1	6	—	—	5	—	—	Y	—	10	0	T	5	6	4	2	9M32M Strela-2M (Grail)
SA-14	1974	M	7	BT	1	6	—	—	4	—	—	Y	—	9	0	T	6	7	4	2	9M34 Strela-3 (Gremlin)
SA-16	1981	M	8	BT/2	1	6	—	—	4	—	—	Y	—	12	0	T	6	7	4*	2*	9K310 Igla-1 (Gimlet)
SA-18	1983	A	8	BT/2	1	6	—	—	2	—	—	Y	—	12	0	T	7	8	4*	2*	9K38 Igla (Grouse)
FIM-43C	1968	M	7	BT	1	6	—	—	5	—	—	Y	—	10	0	T	6	7	4	2	Redeye Block III
FIM-92A	1981	A	8	BT/2	1	2	—	—	3	—	—	Y	—	14	0	T	7	8	5	3	Stinger
FIM-92B	1986	A	8	BT/2	1	2	—	—	1	—	—	Y	—	14	0	T	7	8	5	3	Stinger POST
FIM-92C	1989	A	8	BT/2	1	2	—	—	1	—	—	Y	—	14	0	T	7	8	5	3	Stinger RMP
FIM-92D	1992	A	8	BT/2	1	2	—	—	1	—	—	Y	—	14	0	T	7	8	5	3	Stinger RMP
Blowpipe	1975	OG	8	HT	1	6	—	3	0	—	—	—	—	8	0	T	3	7	6	5	
Javelin	1984	OG	8	ET	1	5	—	2	0	—	—	Y	—	10	0	T	5	8	6	5	Javelin GL
Starburst	1989	LG	8	ET	1	5	—	1	0	—	—	Y	—	10	0	T	5	8	6	5	Javelin S15
Starstreak	2000	LG	8	ET	1	5	—	1	0	—	—	Y	—	24	0	T	7	9	6	4	
Mistral	1990	A	8	ET	1	5	—	—	2	—	—	Y	—	18	0	T	7	9	6	3	
RBS 70	1977	LG	8	ET	1	6	—	2	0	—	—	—	—	12	0	T	5	8	6	3	
Ayn-al-Saqr	1984	I	7	BT	1	6	—	—	5	—	—	Y	—	10	0	T	5	7	4	3	

EWR AND CCU UNITS

An early-warning radar (EWR) can detect aircraft at longer ranges than the target acquisition radars typically associated with SAM units. It can then pass this information down to SAMs in their integrated air-defense system, sometimes with the intermediation of a CCU, to improve the chance of a target acquisition and lock-on.

A command-and-control unit (CCU) acts to coordinate multiple independent SAM firing units in an integrated air-defense system. For early SAMs, where each battery can engage one target, CCUs are typically at the regimental level (i.e., the formation above the battery). For later SAMs, where each launcher can engage one or more targets, they are typically at the battery level.

Figure **??** shows the representation of EWR and CCU units as counters, and Table **??** summarizes their properties.

Properties: In addition to the properties common to other ground units, Table **??** gives: the EWR class, its frequency band, and its range in mega-hexes.

EWR-A:

EWR-B: